

# Linkai Peng

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## TECHNICAL SKILLS

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**Research interest:** Computer Vision, Vision Language Model, Multimodal, Medical Image Analysis

**Programming:** Python (Pytorch, Tensorflow/Keras, OpenCV, Scikit-Image, Sklearn, Matplotlib, Numpy)

**Toolkits:** CUDA, MONAI, nnUNet, SimpleITK, AWS (EC2, S3), SQL, Docker

## EDUCATION

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09/2023 – Present *Northwestern University*

Ph.D. in Electrical Engineering, advisor: Prof. Ulas Bagci (<https://bagcilab.com/>)

09/2019 – 06/2023 *Southern University of Science and Technology*

B.Eng in Computer Science and Technology, advisor: Prof. Xiaoying Tang

## RESEARCH EXPERIENCES

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### Adaptation of Foundation Models

2026 - Present

- Developed a CT-based **bronchoscopy feasibility prediction** pipeline to assess airway accessibility to pulmonary nodules for clinical decision support.
- Adapted large-scale vision-language models to 3D medical imaging settings using Low-Rank Adaptation (**LoRA**), enabling parameter-efficient fine-tuning under limited annotated data.

### Robust Medical Image Analysis

2026 - Present

- Constructed **CrossPan**, a large-scale benchmark (1,386 scans, 8 centers) to evaluate cross-sequence robustness in pancreas MRI.
- Conducted controlled **cross-sequence** and **cross-center** evaluations, demonstrating that physics-driven contrast variations across MRI sequences constitute the dominant source of performance degradation.

### Multi-modal Segmentation

2025 - Present

- Developed a **3D tumor-adaptive framework** (TAGS) that integrates **multi-level alignment** adapters, dual-category CLIP-based **text prompts**, and annotation-free **organ guidance** to extend 2D SAM for volumetric tumor segmentation.
- Achieved **superior performance** on liver, kidney, and pancreas tumor benchmarks, highlighting improved volumetric consistency and robustness compared to prior SAM-based and prompt-driven baselines.
- Designed a **text-guided 3D prostate lesion segmentation framework**. Integrated **BiomedCLIP** with a foreground-only alignment loss and confidence-gated refinement.
- Improved **lesion localization** and **boundary precision** on the PI-CAI benchmark.

## MAJOR PUBLICATIONS

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- 2026 **Peng, L.**, Sun, C., Wang, B., Rowell, J., Gao, C., Aktas, H. E., ... & Bagci, U. (2026). Anatomy-Aware Prediction of Bronchoscopic Accessibility from 3D CT. (accepted in **MICCAI 2026**)
- 2026 Sun, C., **Peng, L.**, Murphy, A., Keles, E., Patel, H., Ross, A., ... & Bagci, U. (2026). Align then Refine: Text-Guided 3D Prostate Lesion Segmentation. (accepted in **EMBC 2026**)
- 2026 **Peng, L.**, Sun, C., Zhang, Z., Dou, W., Aktas, H. E., Bejar, A. M., ... & Bagci, U. (2026). CrossPan: A Comprehensive Benchmark for Cross-Sequence Pancreas MRI Segmentation and Generalization. In Medical Imaging with Deep Learning-Validation Papers. (accepted in **MIDL 2026**)
- 2025 Li, S., **Peng, L.**, Zhang, Z., Durak, G., & Bagci, U. (2025). TAGS: 3D Tumor-Adaptive Guidance for SAM. arXiv preprint arXiv:2505.17096. (accepted in **ICCV-APAH 2025**)